Examiners: Dr. Samir Kamal & Dr. Sara Hassan

Level- 4, Computer Eng. & IT Tech. De

Time: 60 Min

Number of Pages: one

Number of Questions: 2

Attempt all questions

Question 1 [20 Points]

a- In sampling process, drive the sampling condition.

b- In A/D, in detail, explain the quantization process of continuous-amplitude signals.

c- Consider the following analog sinusoid signal: $x(t) = 3\sin(100\pi t)$

(i) Sketch the signal for [0, 30] ms.

(ii) The signal $x(t)$ is sampled with sampling rate Fs=300 samples per second. Determine the frequency of the discrete-time signal X(n) and show that it is periodic.

(iii) Compute the sample values in one period of X(n). Sketch X(n) on the same diagram with $x(t)$.

Question 2 [20 Points]

a- Find the relationship between the frequency variable f for analog signals and the frequency variable F for discrete-time signals and the sampling frequency F_s.

b- Prove that the DFT, X(K), of a sequence x(n) is periodic

c- For the following sequence: x(n) = {1, 0, 1, 1}:

(i) Using the DFT method find the frequency components of x(n), X(k),

(ii) Check your results, obtained in part (i), by calculating x(n) using the IFFT algorithm.

(iii) Using your results, verify Parseval's theorem.

(iv) Plot the amplitude and the phase spectrums.

(v) What is the computational saving when using FFT algorithm instead of DFT algorithm?